M. BASHEER ALI

STORAGE QA ENGINEER
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CAREER OBJECTIVE

Storage QA Engineer with 2.9 years of experience in testing enterprise storage arrays, servers, and drives. Skilled in Python-based automation, performance benchmarking, and validating storage protocols including FC, iSCSI, and NVMe. Experienced in debugging complex storage issues and working across Linux, VMware, and Windows environments. Adept at designing and executing test strategies to ensure high availability, reliability, and performance of storage solutions.

TECHNICAL SKILLS

Storage Management: DAS, SAN, Storage Arrays
Storage Technologies: HDD, SSD, SATA, SAS, iSCSI, RAID, NVMe
Storage & Testing Tools: IOmeter, FIO, mdadm, stocli, smartctl, IPMI, JIRA
Programming & Automation: Python, Pytest
Testing: Manual & Automation Testing, SDLC, STLC, Agile Methodology, Bug Life Cycle, Bug Tracking
Servers & Virtualization: BIOS, BMC, CPLD, VM's, Snapshots, Cloning, HA, FT, DRS
Operating Systems: RHEL 9.2, Windows (2019, 2022), VMWare ESXI (7.0, 8.0)
Version Control: Git, Bitbucket

PROFESSIONAL SUMMARY

- Proficient in Python programming with strong knowledge of data types, functions, file handling, regular expressions, and OOP concepts including class, object, inheritance, polymorphism, and encapsulation
- Experienced in developing automation scripts using Python and Pytest framework
- Strong understanding of software testing methodologies including Functional Testing, Re-testing, Regression Testing, and Performance Testing
- Familiar with testing techniques such as Black Box, White Box, and Gray Box Testing
- Good knowledge of SDLC, STLC, Bug Life Cycle, and Agile methodology
- Experience in writing, modifying, and executing test cases and automation scripts
- Good knowledge of storage devices including HDD, SSD, Storage Arrays, and JBOD
- Skilled in analyzing storage performance using IOPS, latency, and throughput metrics with tools like FIO and IOmeter
- Experienced in identifying disk failures using SMART data and system logs
- Knowledge of SSD concepts including wear leveling, write amplification, garbage collection, and over-provisioning
- Strong understanding of RAID concepts and features such as Hot Spare, ORLM, Capacity Expansion, Consistency Check, and Hot Swapping
- Familiar with storage architectures including DAS, SAN, and NAS and protocols such as SATA, SAS, SCSI, and NVMe
- Hands-on experience with tools like PyCharm, JIRA, and PuTTY
- Comfortable working in Linux, Windows, and VMware environments
- Knowledge of x86 server architecture and firmware updates including BIOS, BMC, and CPLD

- Familiar with virtualization concepts such as vMotion, Storage vMotion, High Availability (HA), Fault Tolerance (FT), DRS, Snapshots, Cloning, and Templates

PROFESSIONAL EXPERIENCE

Company Name – X company Oct 2023 - present
Role: Storage QA Engineer
Client – DDN (DataDirect Networks)

Project1: DDN IntelliFlash – Platform & Feature Validation (Oct 2023 – Sep 2025)

Project Description

Worked on the validation of the DDN IntelliFlash enterprise all-flash storage platform, ensuring hardware, firmware, and software features met quality, performance, reliability, and stability requirements. Performed platform and feature validation across different storage configurations, firmware versions, and enterprise deployment scenarios. Executed functional, regression, compatibility, and performance testing while identifying defects and collaborating with cross-functional teams to deliver high-quality storage solutions.

Roles & Responsibilities

- Performed platform and feature validation of DDN IntelliFlash enterprise storage systems, ensuring functionality, stability, reliability, and high availability.
- Validated storage components including controllers, SSD/HDD drives, HBAs, network interfaces, power supplies, and storage enclosures across hardware and firmware configurations.
- Executed RAID, storage pool, volume/LUN, hot-spare, degraded-mode, rebuild, and recovery validation under various I/O workloads.
- Conducted FC and iSCSI host connectivity, multipathing, path failover, controller failover/failback, reboot, and power-cycle validation while maintaining continuous I/O.
- Performed firmware upgrade/downgrade, compatibility, fault-injection, regression, and system recovery testing to ensure platform resiliency.
- Conducted performance, stress, and endurance testing with FIO, evaluating IOPS, throughput, latency, block sizes, queue depths, and read/write workload patterns.
- Developed Python/Pytest automation leveraging Paramiko/SSH for remote test execution, storage health checks, log collection, result validation, and repetitive workflow automation.
- Leveraged AI-assisted tools such as ChatGPT/OpenAI and GitHub Copilot for automation development, code optimization, test-scenario generation, troubleshooting, and intelligent log analysis.
- Analyzed controller, drive, OS, SMART, multipath, and performance logs, incorporating AI-assisted analysis to identify error patterns, performance anomalies, and potential root causes.
- Integrated automated validation with Jenkins and Git, and managed defect reporting, evidence collection, root-cause investigation, and regression verification through Jira.

Project2: DDN A3I – Drive Validation (Sep 2025 – present)

Client: DDN (DataDirect Networks)

Project Description

Performed enterprise drive validation for the DDN A3I storage platform, focusing on HDD, SSD, and NVMe qualification, compatibility, performance, reliability, firmware, and fault-recovery testing. Developed Python-based automation and leveraged AI-assisted tools for test development, log analysis, troubleshooting, and validation efficiency.

Roles & Responsibilities

- Executed end-to-end qualification and validation of enterprise HDD, SSD, and NVMe drives on the DDN A3I storage platform.
- Validated drive detection, capacity, firmware, interface speed, health status, compatibility, and behavior across supported configurations.
- Conducted performance benchmarking with FIO, analyzing IOPS, throughput, latency, and bandwidth across sequential, random, and mixed workloads.
- Performed stress, endurance, power-cycle, hot-plug, fault-injection, and long-duration I/O testing to assess drive reliability and stability.
- Validated RAID behavior during drive failures, degraded states, hot-spare activation, rebuild operations, and recovery scenarios.
- Executed drive firmware upgrade and downgrade validation, including compatibility, stability, and post-upgrade data-integrity verification.
- Diagnosed drive and I/O issues through SMART data, nvme-cli, storcli, dmesg, and Linux system/storage logs.
- Developed Python, Pytest, and Paramiko automation for drive discovery, health monitoring, FIO execution, result parsing, log collection, and regression testing.
- Leveraged ChatGPT and GitHub Copilot for automation development, test-scenario generation, log-pattern analysis, debugging, and optimization of storage validation workflows.
- Documented defects in Jira with reproducible steps, logs, test evidence, and impact analysis, while collaborating with development and firmware teams for root-cause analysis and fix verification.

EDUCATION

Bachelor of Science (Computer Science) | PRGC(A)**.** **2020 – 2023**

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

M. Basheer Ali